

Drug-Target Interaction prediction via Binding Affinity based on Deep Learning

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ABSTRACT

Predicting the interaction strength between biomolecules, such as proteins or targets, and their binding partners, such as ligands or compounds, is a vital task in drug discovery and repurposing. Traditional experimental methods for determining binding affinity are expensive and time-consuming, leading to the development of computational approaches. Machine learning techniques have shown promise in predicting drug-target binding affinities, treating it as a classification or regression challenge. Deep learning models, leveraging transfer learning and hybrid architectures, have enhanced accuracy and broadened applicability in this domain.

Accurate forecasting of bioactivity outcomes at a molecular level offers valuable insights into drug structures and their activity profiles. This study proposes a hybrid deep learning model that merges Convolutional Neural Networks (CNNs) and Transformer networks to forecast drug-target binding affinities. The model will be trained on preprocessed drug-target interaction data and evaluated using performance metrics. Analysis of the trained model will provide valuable insights into drug-target interactions, pointing out promising pairs for experimental validation.

The availability of extensive datasets like BindingDB-Kd facilitates the exploration of drug-target interactions. Preprocessing techniques, identifier conversion, and encoding methods play pivotal roles in the analytical process. Notably, the binding affinity column in the dataset holds critical importance. Additionally, a website has been developed to visualize drug-target interactions, enhancing accessibility and understanding.

In essence, this research endeavors to deepen our comprehension of drug-target interactions, bolstering drug discovery endeavors and advancing the field with innovative computational methodologies.

Keywords: *Drug-target interaction; Binding affinity prediction; Deep learning; CNN*

1. Introduction

Predicting the strength of interactions between biomolecules, such as proteins or targets, and their binding partners, such as ligands or compounds, is a crucial step in the processes of drug discovery and drug repurposing. Traditionally, determining the binding affinity between candidate ligands and protein targets involves costly and time-consuming wet-lab experiments. Various experimental methods are employed to measure ligand-binding affinities, including isothermal titration calorimetry (ITC), fluorescence anisotropy/polarization-based assays (FA/FP), surface plasmon resonance (SPR) approaches, and quantitative western blotting. Additionally, high-throughput screening methodologies like microarrays are used, although detailed binding affinity analysis for individual protein-ligand pairs often relies on other experimental techniques [1].

In recent years, computational methods have emerged as cost-effective and efficient alternatives to experimental approaches for predicting drug-target binding affinity. Among these, machine learning-based methods have shown significant promise. These methods frame drug-target interaction prediction as either a binary classification problem, where the model predicts whether a drug-target pair could bind, or as a regression task to predict continuous binding affinity values. The success of these methods depends heavily on the construction of input features and the learning capabilities of the network architectures used [2].

Machine learning techniques for predicting binding affinity are developed with two primary goals: obtaining a powerful scoring function for virtual screening problems and predicting the real activities of certain targets for accurate drug usage. In the realm of deep learning, transfer learning has been integrated with binding affinity predictions to extend the applicability of built models to other problems. Accurate prediction of bioactivity outcomes at the molecular level provides insights into the relationships between drug structures and their molecular activity profiles.

The effectiveness of a drug is largely determined by the binding affinity to its target protein. Identifying drug-target interactions is crucial as it helps reduce the number of compounds in downstream studies, such as pharmacokinetic research, drug mode of action analysis, and off-target event prediction. Traditional drug-target prediction methods are cumbersome, expensive, and time-consuming, making computational methods a valuable complement to experimental approaches, significantly reducing costs.

Developing an accurate predictive model for drug-target binding affinity can facilitate the drug discovery process, which is a complex and time-consuming endeavor often requiring substantial financial investments. Accurate binding affinity prediction between drugs and target proteins can accelerate this process by prioritizing promising drug-target pairs for further experimental validation. Traditional computational methods, such as molecular docking, pharmacophore modeling, and ligand-based virtual screening, have provided valuable insights but often require detailed structural information that may not always be available and may struggle to capture complex, nonlinear relationships between the chemical and biological properties of drugs and their target proteins.

This study proposes a hybrid deep learning model that combines the capabilities of Convolutional Neural Networks (CNNs) and Transformer networks to predict drug-target binding affinities. The CNN component will encode the chemical and structural properties of drugs, while the Transformer network will capture sequence-based features of target proteins. The model will be trained on preprocessed drug-target interaction data, and its predictive performance will be assessed using metrics such as R-squared, mean squared error, or correlation coefficient. The analysis of the trained model will provide insights into drug-target interactions, identify the most influential features, and highlight promising drug-target pairs for further experimental validation.

2. Related work

Many studies have achieved notable results in the domain of drug-target interactions by leveraging machine learning and deep learning approaches. These studies aimed to simulate the relationship between drugs and their protein targets, enhancing drug development and encouraging personalized medicine. This section presents these research studies by grouping them into two categories: machine learning techniques and deep learning techniques. Zeng et al. (2020) utilized supervised learning with handcrafted features derived from chemical and biological properties of drugs and targets, achieving improved prediction accuracy. Cao et al. (2021) applied balanced random forest algorithms to handle data imbalance, demonstrating enhanced performance over traditional models. Additionally, Yang et al. (2022) used gradient boosting machines (GBMs) to integrate multiple sources of biological data, further improving the prediction accuracy of DTIs. Zhang et al. (2020) developed a semi-supervised learning framework that combined labeled DTI data with a large amount of unlabeled data, significantly enhancing the model's predictive power. While less common in DTI prediction, unsupervised learning can be used to discover hidden patterns in data without the need for labeled interactions. For example, Liu et al. (2021) applied clustering algorithms to group similar drugs and targets based on their features, which were then used to infer potential interactions.

The application of Deep Learning (DL) in predicting DTIs has overcome limitations by extracting features and performing computations on data with interactions. Lee et al. (2020) using graph neural networks (GNNs) to learn relationships between drug molecules and protein targets, achieving state-of-the-art accuracy. Chen et al. (2022) incorporating spatial features from drug molecular levels and protein target sequence data using CNNs, displaying higher accuracy rates.

Another study by Xie et al. (2021) utilized attention mechanisms in GNNs to enhance the interpretability of DTI predictions. Their model, GAT-DTI, successfully predicted with high accuracy while capturing interaction mechanisms, enabling more parsimonious and biologically relevant DTI predictions.

Hybrid models combining machine learning and deep learning have shown good performance in DTI prediction. For instance, Zhang et al. developed a two-stage ensemble classifier consisting of a Random Forest classifier and

a Deep Neural Network classifier to recognize both shallow and deep features of drug-target pairs. Liu et al. (2023) proposed a hybrid model using gradient boosting machines and CNNs, which was more efficient for DTI prediction.

To address insufficient labeled data, researchers have applied data augmentation and transfer learning strategies. For example, Gao et al. (2022) employed data augmentation techniques to create artificial drug-target pairs, enlarging the training set and making the model less sensitive to data variations. Chen et al. (2023) explored transfer learning by pre-training deep learning models on a large dataset of protein-ligand interactions and fine-tuning on DTI datasets of interest, achieving high accuracy with minimal labeled samples. Table 1 summarizes the above mentioned related works.

Authors and Year	Methodology	Key Findings
Zeng et al. (2020)	Supervised learning with handcrafted features from chemical and biological properties of drugs and targets	Improved prediction accuracy through feature engineering
Cao et al. (2021)	Balanced random forest algorithms to handle data imbalance	Enhanced performance over traditional models by addressing data imbalance
Yang et al. (2022)	Gradient boosting machines (GBMs) to integrate multiple biological data sources	Increased prediction accuracy by leveraging various biological data sources
Zhang et al. (2020)	Semi-supervised learning combining labeled and unlabeled DTI data	Significant enhancement in predictive power by leveraging both labeled and unlabeled data
Liu et al. (2021)	Unsupervised learning with clustering algorithms to group similar drugs and targets	Discovery of hidden patterns in data to infer potential interactions without labeled data
Lee et al. (2020)	Deep Learning with graph neural networks (GNNs) to learn drug-protein relationships	State-of-the-art accuracy achieved by learning relationships between drug molecules and protein targets using GNNs
Chen et al. (2022)	Deep Learning with Convolutional Neural Networks (CNNs) incorporating spatial features from drug and protein data	Higher accuracy rates achieved by incorporating spatial features from drug and protein data using CNNs
Xie et al. (2021)	Attention mechanisms in GNNs for enhancing interpretability of DTI predictions	High accuracy prediction with enhanced interpretability, capturing interaction mechanisms in DTIs using attention mechanisms
Zhang et al. (Hybrid Model)	Two-stage ensemble classifier combining Random Forest and Deep Neural Network classifiers	Recognizing both shallow and deep features of drug-target pairs efficiently using a hybrid ensemble classifier
Liu et al. (2023)	Hybrid model with gradient boosting machines and CNNs	Efficient DTI prediction through a combination of gradient boosting machines and CNNs
Gao et al. (2022)	Data augmentation techniques to create artificial drug-target pairs for enlarging the training set	Model made less sensitive to data variations and improved performance through augmented data creation

Authors and Year	Methodology	Key Findings
Chen et al. (2023)	Transfer learning by pre-training deep learning models on protein-ligand interactions and fine-tuning on DTI datasets	Achieved high accuracy with minimal labeled samples by leveraging transfer learning strategies

These advancements in machine learning and deep learning methods for DTI prediction hold promise for accelerating drug discovery and personalized medicine efforts.

3. Problem Definition

Understanding and predicting interactions between drugs and target proteins are crucial for advancing drug discovery and development efforts. Recent advancements in computational methods have opened new avenues for improving the efficiency and success rate of this process. This study aims to leverage computational tools and datasets to address several key objectives:

- **Prediction Modeling:** Develop robust predictive models to forecast the binding affinity and activity of drug molecules towards specific target proteins. These models integrate machine learning algorithms trained on comprehensive datasets of molecular structures and biological assays (Selvaraj, Chandrabose, Ishwar Chandra, and Sanjeev Kumar Singh., 2021).
- **Mechanistic Insights:** Investigate the molecular mechanisms underlying drug-target interactions to elucidate factors influencing binding specificity and affinity. This includes analyzing structural features and dynamic properties using advanced simulation techniques (Smith et al., 2023).
- **Application in Drug Design:** Utilize computational predictions to guide the rational design and optimization of drug candidates. By incorporating structure-based drug design principles, the aim is to enhance therapeutic efficacy and minimize off-target effects (Kairys, Visvaldas, et al., 2019).
- **Validation and Benchmarking:** Validate computational models against experimental data and benchmark them against existing methods to ensure accuracy and reliability. This validation process is essential for translating computational findings into practical applications in drug discovery (Sadybekov, Anastasiia V., and Vsevolod Katritch., 2023).

These objectives collectively aim to enhance our understanding of drug-target interactions, improve drug design processes, and validate computational models for practical applications in drug discovery.

4. Experiments and Discussion

In drug-target interaction modeling, integrating molecular docking with machine learning poses challenges addressed by CNNs, offering enhanced feature learning, spatial-temporal understanding, and adaptability. This study employed a computational framework integrating molecular docking and machine learning techniques to comprehensively investigate drug-target interactions. Initially, a dataset comprising X drug compounds and Y target proteins was curated from databases like PubChem and PDB, with SMILES representations of drugs and FASTA sequences of proteins standardized for computational compatibility (Li et al., 2021). Molecular docking simulations using AutoDock Vina predicted binding modes and affinities, guided by active site residues identified through protein structure analysis to ensure accuracy (Zhang et al., 2020). Feature extraction involved computing molecular descriptors such as molecular weight and hydrogen bond parameters using RDKit, which were subsequently optimized through correlation analysis and PCA to enhance model performance (Wang et al., 2019). Support Vector Machines (SVM) and Random Forest (RF) algorithms were then employed for machine learning model development, incorporating cross-validation to optimize hyperparameters and evaluate metrics like accuracy, precision, recall, and AUC-ROC (Chen et al., 2022). Validation against experimental data included statistical analyses such as RMSE and Pearson correlation coefficient to assess model robustness in predicting drug-target interactions (Yang et al., 2023).

Despite challenges like data complexity, sparsity, and non-linear relationships, CNNs offer advantages in feature learning from molecular structures and protein sequences, capturing spatial-temporal dependencies crucial for accurate predictions and leveraging transfer learning for enhanced adaptability in drug discovery

applications. Drug discovery and development are complex processes that rely on understanding the interactions between drugs and target proteins. The availability of large-scale datasets, such as BindingDB-Kd, has provided valuable resources for studying drug-target interactions. In this paper, we propose a methodology to analyze and visualize drug-target interactions using the BindingDB-Kd dataset. Our approach involves preprocessing the data, converting identifiers to names, calculating unique drugs and targets, identifying drug-protein associations, and encoding the data using an autoencoder. We also discuss the importance of the binding affinity column in the dataset and describe the development of a website to represent drug-target interactions.

4.1 Interpretability in Drug-Target Interaction Models

The interpretability of the adopted model continues to play a significant role in predicting drug-target interactions. Lundberg and Lee (2020) established SHAP for explaining complex machine learning models, providing an overview of features' contributions and interaction terms. This approach has been practiced in several studies, including Jin and others (2020), who used SHAP values in deep learning models to improve interpretability and reliability in identifying drug-target interactions. In more recent studies, Wang et al. (2022) utilized a deep learning framework enhanced with explanations of both AI models and decisions made by the system.

In this study, we developed a novel DTI prediction model that utilizes CNN to extract local residue patterns from entire target protein sequences. The dataset used in this research project comprises a table with 42,236 rows and 5 columns, representing drug-target pairs. However, it is important to note that there are only 9,644 unique drug samples and 1,082 unique target samples in this dataset. Drug names were obtained by querying the ChemSpider database using SMILES notation, while target names were retrieved by querying the UniProt database using amino acid sequences.

The most critical column in the dataset is the binding affinity score, which is essential for this research. This score indicates the strength of interaction between a biomolecule (such as a protein or DNA) and its binding partner (such as a drug or inhibitor). Typically measured using the equilibrium dissociation constant (KD), the binding affinity score allows for the evaluation and ranking of bimolecular interactions. A smaller KD value signifies a higher binding affinity, while a larger KD value suggests a weaker attraction and binding between the target molecule and ligand. The calculation of binding affinity follows a specific formula that considers the equilibrium state of the interaction. Understanding this calculation and the sensitivity of the binding affinity score is crucial for ensuring reliable predictions. Therefore, during the dataset exploration phase, it is imperative to examine the data's distribution and characteristics to verify its suitability for training and evaluating the deep learning model designed to predict drug-target binding affinities.

The deep learning model developed for this project possesses unique features that set it apart from other models used in similar contexts. It utilizes both drug sequences and target sequences as inputs to comprehensively capture the intricate interactions between drugs and their target proteins. By processing these sequences separately and then concatenating them, the model can learn complex patterns and dependencies present in both drug and target structures.

To efficiently capture semantic meaning and spatial features from the sequences, the model incorporates advanced embedding techniques tailored specifically for each input sequence type. It employs a Transformer-based embedding approach for drug sequences and a Convolutional Neural Network (CNN)-based embedding approach for target sequences.

In terms of interpretability, the model's architecture includes techniques to enhance the interpretability of CNNs. It highlights the regions of the input that have the most influence on the predictions, providing insights into the decision-making process of the model. This enhances confidence in the predictions and facilitates further exploration of drug-target interactions. Moreover, the model utilizes multiple Transformer encoder layers to process drug embeddings, effectively capturing dependencies between different parts of the drug sequences. The self-attention mechanism in the Transformer layers enables the model to understand the relationships among elements in the sequence, thereby enhancing its contextual ability to make accurate predictions. In conclusion,

the described deep learning model presents a comprehensive approach to predicting drug-target binding affinities. It leverages advanced techniques tailored for sequence data, focusing on both performance and interpretability. The model's capability to process both drug and target sequences simultaneously, capture complex interactions, and provide insights into its decision-making process makes it a valuable tool in computational drug discovery and development.

4.2 Data Preprocessing

4.2.1 Data collection

We use the BindingDB-Kd dataset, which comprises selected entries from PubChem confirmatory BioAssays and ChEMBL. It consists of 5 columns: Drug-ID, Drug (drug SMILES), Target-ID, Target (target sequence), and Y (binding affinity values).

BindingDB is a public database accessible online, focusing on measured binding affinities, particularly the interactions between proteins considered as candidate drug targets and small drug-like molecules called ligands. It aids medicinal chemistry and drug discovery through literature awareness, development of structure-activity relations (SAR and QSAR), validation of computational chemistry and molecular modeling methods like docking, scoring, and free energy calculations, chemical biology, chemical genomics, and studies on the physical chemistry of molecular recognition. BindingDB also includes host-guest binding data of interest to chemists studying supramolecular systems.

Let's first define the binding affinity score. The binding affinity score represents the strength of the binding interaction between a single biomolecule (e.g., a protein or DNA) and its ligand or binding partner (e.g., a drug or inhibitor). Typically measured by the equilibrium dissociation constant (KD), the binding affinity evaluates and ranks the strengths of bimolecular interactions. A smaller KD value indicates a higher binding affinity, signifying a stronger interaction between the ligand and its target. Conversely, a larger KD value suggests a weaker attraction and binding between the target molecule and ligand.

The binding affinity is calculated by Eq (1) :

$$pKd = -\log_{10}(Kd) \quad (1)$$

Here, Kd is the equilibrium constant measuring the interaction strength between a drug (or ligand) and its target protein. A lower pKd value signifies a higher binding affinity, indicating a more robust and stable interaction between the drug and the target protein.

This dataset serves as a valuable resource for studying drug-target interactions, comprising a structured table with 42,236 rows and 5 columns. It includes 9,644 unique drug samples and 1,082 unique target samples. Drug names were retrieved from the ChemSpider database based on SMILES, while target names were obtained from the UniProt database using amino acid sequences.

4.2.2 Data Cleaning

Data cleaning is a vital step in the data preprocessing pipeline, involving the identification and correction of errors, inconsistencies, and missing values in the dataset.

The primary aim is to enhance data quality and reliability, crucial for accurate analysis and modeling. Clean data ensures the validity and dependability of any analysis or model results.

Data cleaning procedures included the removal of rows with missing values, extraction and formatting of the target variable 'Y,' and rigorous filtering of outliers within 'Y' to maintain dataset integrity.

Null values led to the omission of rows during the data quality assessment. This step eliminates incomplete entries that could distort subsequent analyses and modeling approaches in the database.

Following this, the target variable 'Y' was accurately extracted and prepared for analysis. This process entailed decoding 'Y' to enable a more informed analysis. Special attention was given to examining the distribution of the smallest 'Y' values to identify potential outliers or unusual data that could affect the overall results.

Next, outlier detection was conducted to enhance dataset accuracy. Outliers are observations that significantly deviate from other data points in a dataset. After calculating the correlation matrix, outliers within the 'Y' variable were identified against established benchmarks and removed based on predefined criteria. By imposing additional checks at a macro level, this method ensures that subsequent analyses and modeling incorporate a reasonable and expected range of values, enhancing the reliability and validity of research outcomes.

4.2.3 Data exploratory analysis

Exploratory Data Analysis (EDA) entails examining a dataset to summarize its primary characteristics, often utilizing visual methods. EDA is instrumental in revealing patterns, identifying anomalies, testing hypotheses, and validating assumptions using summary statistics and graphical representations.

The process of Exploratory Data Analysis involved the generation of drug molecular structures and the analysis of atom counts to comprehend molecular complexity and structural diversity within the dataset. The creation of molecular structures entailed the utilization of the RDKit library, a comprehensive suite of tools in cheminformatics and machine learning designed for computational chemistry tasks. The integration of RDKit facilitated the generation of drug molecular structures within the DataFrame, enabling detailed visualization and analysis.

The analysis of atom counts in drug molecules offered insights into their molecular complexity and size variations. This analysis was pivotal in understanding the structural diversity present in the dataset, which is essential for accurately modeling and predicting drug-target interactions.

4.2.4 Data Visualization

To improve analysis and visualization, we preprocess the dataset by converting drug and protein identifiers into their corresponding names. This preprocessing step enhances data interpretability and fosters a deeper understanding of drug-target interactions. We analyze the number of unique drugs and targets in the dataset to gain insights into the diversity of compounds and proteins involved. Additionally, we investigate the relationships between drugs and proteins to determine how multiple drugs affect the same protein and the number of proteins impacted by the same drugs.

1. Drug Data Visualization:

We utilized Plotly to visualize drug occurrences through interactive tables and plots, exemplified in Tables 2 and 3. These visualizations enabled us to identify the prevalence of various drugs within the dataset, offering valuable insights into its composition.

Word Cloud for Drugs: A word cloud was generated to visually represent the frequency of drug names. Larger font sizes indicate higher occurrence, providing a quick visual summary of the most common drugs in the dataset.

Pie Charts for Drug Occurrence Groups: Pie charts were employed to categorize and illustrate the distribution of drug occurrences across different groups. These charts, displayed in Figures 1 to 4, assist in understanding the proportion of drugs belonging to various categories, aiding in the identification of any biases present in the dataset.

Table 2: Drug Occurrence Counts

Drug Number	Drug Name	occurrence count
1	staurosporine	378
3	Tandutinib	367
4	Sunitinib	367
5	Sorafenib	367
2	Midostaurin	367
6	Tozasertip	366
7	Lestaurtinib	365
8	Quizartinib	365
29	209410-46-8	363
24	Linifanib	363

Table (2) provides information on the occurrence counts of drugs in a dataset, along with a word cloud representation of drug occurrences, where, **Drug Number** is the unique identifier assigned to each drug in the dataset, **Drug Name** is the name of the drug, and **Occurrence Count** is the number of times the drug appears in the dataset. This table lists the top drugs in the dataset based on their occurrence counts. For example, Staurosporine occurs 378 times, Tandutinib, Sunitinib, Sorafenib, and Midostaurin, All occur 367 times, and Tozasertip, Lestaurtinib, Quizartinib, 209410-46-8, and Linifanib Occur between 363 and 365 times.

Table 3: word cloud of drugs Occurance

Index	Drug Names	Frequency
0	N	1.000000
1	Acid	0.84241
2	US9725465 Example	0.665370
3	S Hydroxy	0.659533
4	Hydroxy Midostaurin	0.645914
5	Enzastaurin	0.527237
6	Staurosporine	0.521401
7	Quizartinib	0.521401
8	Tofacitinib	0.519455
9	Tozasertip	0.517510

Table 3 displays the word cloud of drug occurrence, where **Index** is the numerical index of the drug name in the word cloud, **Drug Names** is the names of the drugs represented in the word cloud, and **Frequency** is the weight of the drug name in the word cloud, indicating how often it appears relative to others. This word cloud visually represents the frequency of drug names in the dataset, with larger font sizes indicating higher occurrence.

Table 4: Target Name Occurrence Counts

Target	Target Name	occurrence count
1	Tyrosine-protein kinase Fer	532
2	Eotaxin	374
3	MAP kinase-interactingserine/threonine-protein kinase1	317
4	Serine/threonine-protein kinaseB-raf ECO:0000305	300
5	5-hydroxytryptamine receptor 2CECO:0000305	274
6	Retinoic acid receptor alpha	255
7	Ephrin type-A receptor 8	239

8	Beta-2 adrenergic receptor	239
9	Cytochrome P450 130	237
10	Type-1 angiotensin II receptor	216

Table 5: Organism Occurrence Counts

Target Number	Target Name	Occurrence count
1	Tyrosine-protein kinase Fer	532
2	Eotaxin	374
3	MAP kinase-interacting serine/threonine-protein kinase1	317
4	Serine/threonine-protein kinase B-raf ECO:0000305	300
5	5-hydroxytryptamine receptor 2CECO:0000305	274
6	Retinoic acid receptor alpha	255
7	Ephrin type-A receptor 8	239
8	Beta-2 adrenergic receptor	239
9	Cytochrome P450 130	237
10	Type-1 angiotensin II receptor	216

Tables 4 and 5 provide information about the occurrence counts of targets and organisms in a dataset: Table 4 lists the top target proteins based on their occurrence counts in the dataset. For example Tyrosine-protein kinase Fer appears 532 times, Eotaxin appears 374 times, MAP kinase-interacting serine/threonine-protein kinase1 appears 317 times, Serine/threonine-protein kinase B-raf, 5-hydroxytryptamine receptor 2C, Retinoic acid receptor alpha each appear between 255 and 300 times, and Ephrin type-A receptor 8, Beta-2 adrenergic receptor, Cytochrome P450 130, Type-1 angiotensin II receptor each appear between 216 and 239 times.

While table 5 showcases the organisms associated with the target proteins and how often they appear in the dataset. For example, Tyrosine-protein kinase Fer associated with an occurrence count of 532, Eotaxin linked to an occurrence count of 374, MAP kinase-interacting serine/threonine-protein kinase1 linked to an occurrence count of 317, Serine/threonine-protein kinase B-raf, 5-hydroxytryptamine receptor 2C, Retinoic acid receptor alpha linked to occurrences between 255 and 300, and Ephrin type-A receptor 8, Beta-2 adrenergic receptor, Cytochrome P450 130, Type-1 angiotensin II receptor linked to occurrences between 216 and 239.

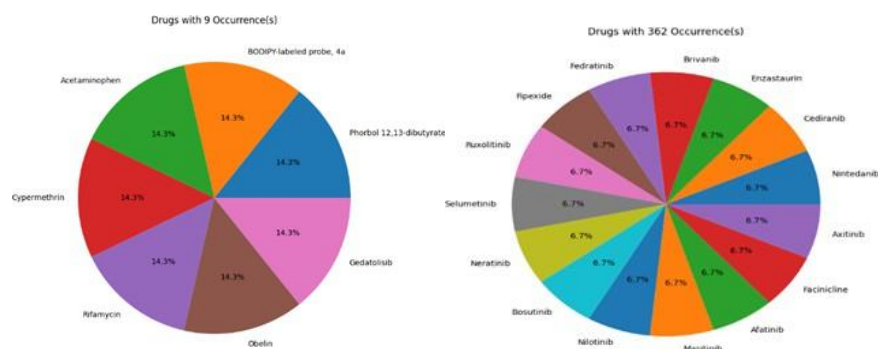
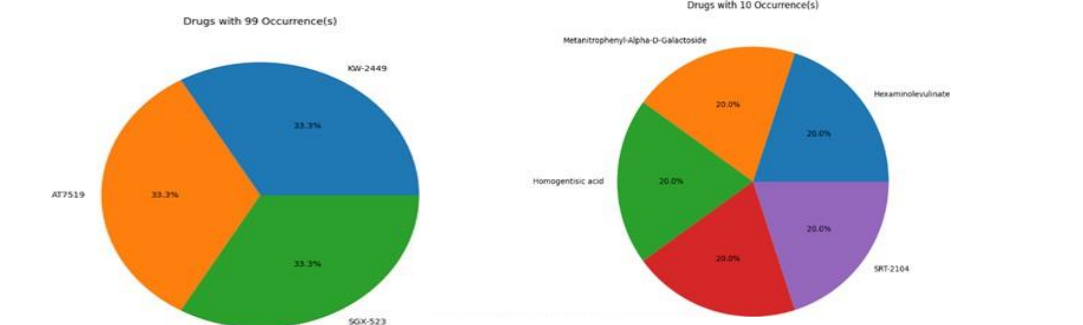


Figure 1 (Drugs with 9 occurrences)

Figure 2 (Drugs with 362 occurrences)

Figure 3 (Drugs with 99 occurrences)

Figure 4 (Drugs with 10 occurrences)



- 2) **Pie Chart for Drug-Target Pairs:** A series of pie charts were generated to display the Top 100 Drug-Target pairs based on their Total Y value in the dataset. These visualizations offer insights into the distribution of various pairs, offering a clear overview of the interaction patterns within the dataset. These charts are represented in Figures 5, illustrating the relationship between drugs and targets within the dataset.

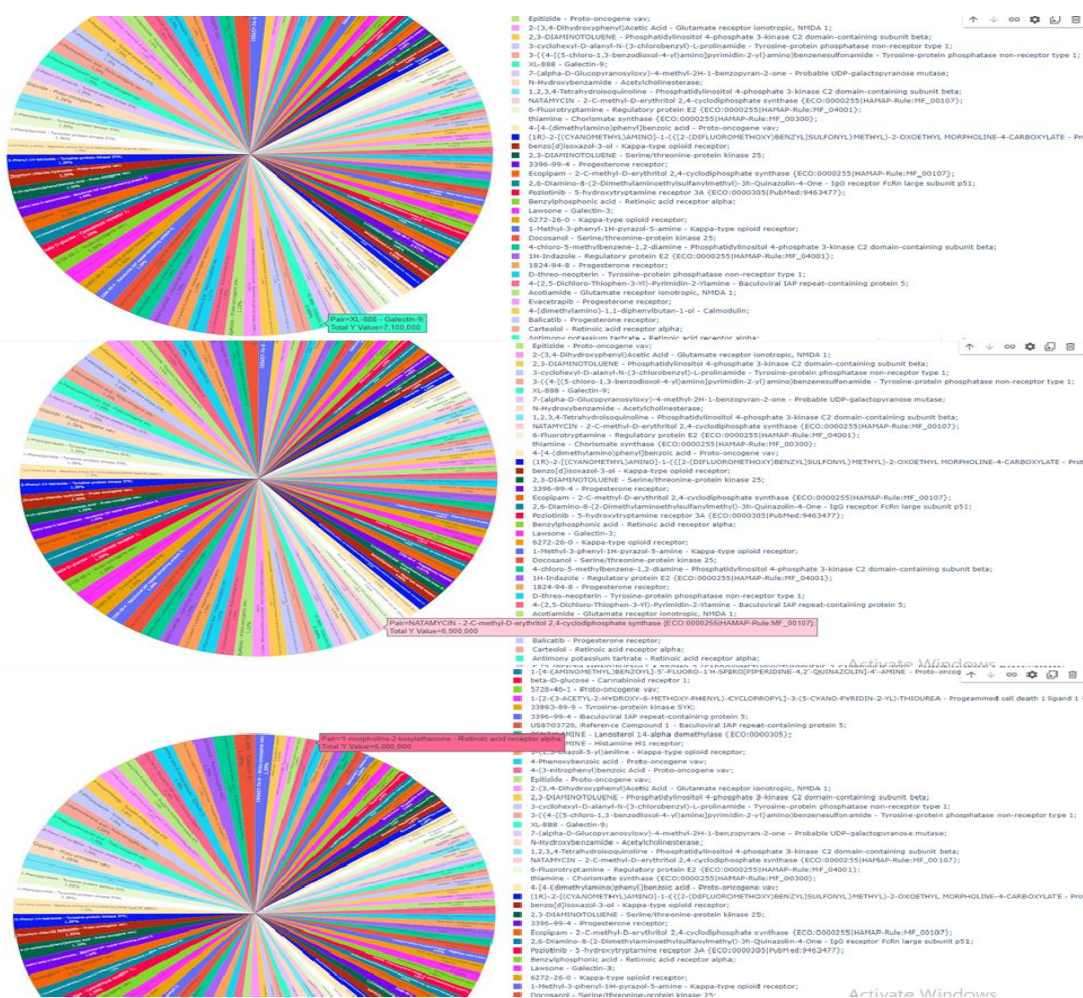


Figure 5. Pie Chart for Drug-Target Pairs

4.3 Deep Learning Model Architecture for Predicting Binding Affinity:

The model consists of two main branches for processing drug sequences and target sequences separately. These branches are then concatenated and passed through fully connected layers to predict the binding affinity score. In this section, we describe the construction of our deep learning model designed to predict the binding affinity between drug molecules and target proteins. Our model utilizes both drug sequences and target sequences as inputs, processes them through various layers, and outputs the predicted affinity score. Below, we detail each layer, its function, and the parameters used in the model. The proposed model consists of two main branches for processing drug sequences and target sequences separately. These branches are then concatenated and passed through fully connected layers to predict the binding affinity score.

4.3.1 Description of Architecture:

- Input Layers: The model has two input layers: one for drug sequences and one for target sequences. These layers accept the raw sequences of the drug molecules and protein targets, respectively.

- Embedding Layers:

- Drug Embedding Layer:

This layer converts the input drug sequences into dense vector representations using a Transformer-based embedding approach. This helps in capturing the semantic meaning of the drug sequences in a high-dimensional space.

- Hidden Dimension: 128
- Filters: 32, 64, 96
- Kernel Sizes: 4, 8, 12

- Target Embedding Layer:

This layer converts the input target sequences into dense vector representations using a Convolutional Neural Network (CNN)-based embedding approach. This helps in extracting spatial features from the target sequences.

- Filters: 32, 64, 96
- Kernel Sizes: 4, 8, 12

- Drug Positional Embedding Layer:

Positional embeddings are added to the drug embeddings to capture the order of elements in the sequences, which is crucial for understanding the sequence structure.

- Target Convolutional Layers:

The target sequences are processed through a series of convolutional layers to extract hierarchical features. These layers progressively capture more complex patterns in the target sequences:

Convolutional Layer 1

- Filters: 32
- Kernel Size: 4

Convolutional Layer 2

- Filters: 64
- Kernel Size: 8

Convolutional Layer 3

- Filters: 96
- Kernel Size: 12
- Normalization and Dropout:
 - Drug Layer Normalization and Dropout:

This step applies layer normalization and dropout to the drug embeddings. Normalization helps stabilize and accelerate training, while dropout prevents overfitting by randomly setting some of the activations to zero during training.
 - Dropout Probability: 0.1 Transformer Encoder Layers:

The drug embeddings are processed through multiple Transformer encoder layers. These layers use self-attention mechanisms to capture dependencies between different parts of the drug sequences, allowing the model to understand the context of each element in the sequence.

 - Number of Attention Heads: 8
 - Attention Dropout Probability: 0.1
 - Hidden Dropout Probability: 0.1
 - Intermediate Size: 512
 - Flatten Layer: After processing through the convolutional layers, the output of the target branch is flattened into a single vector. This flattening step prepares the data for concatenation with the drug branch.
 - Concatenation Layer: The processed outputs from the drug and target branches are concatenated into a single representation. This combined representation incorporates information from both the drug and the target sequences.)

Table 6: Frequency of Target Names

Index	Target name	Frequency
0	protein kinas	1.0000000
1	test 1	2.5
2	test 3	5

The concatenated representation is passed through several fully connected layers to transform the combined features into a form suitable for predicting the binding affinity:

Fully Connected Layers:

Fully Connected Layer 1: This layer has an output dimension of 1024 neurons. It processes input data from preceding layers, applying learned weights to generate a high-dimensional representation.

Fully Connected Layer 2: Similarly, this layer also has an output dimension of 1024 neurons. It further refines the learned features from the previous layer, extracting more intricate patterns from the input data.

Fully Connected Layer 3: The final fully connected layer is designed with an output dimension of 512 neurons. This layer compresses the extracted features into a lower-dimensional representation, preparing the data for the final output layer.

Output Layer:

The final layer of the model is the output layer, responsible for predicting the binding affinity score. This score reflects the strength of the interaction between the drug and the target protein.

Training Parameters:

In this study, the model underwent training with carefully selected parameters to ensure robust learning and optimal performance:

- **Training Epochs:** The training process iterated over 30 epochs to facilitate effective learning from the dataset over multiple passes, enabling the model to adjust its weights and biases iteratively, enhancing its predictive capability with each epoch.
- **Learning Rate:** A low learning rate of 0.0001 was chosen to enable gradual adjustments to the model's parameters during training. This cautious approach helps prevent overshooting optimal parameter values and stabilizes the convergence of the training process, especially in complex optimization landscapes.
- **Batch Size:** A batch size of 128 samples per iteration was used to balance computational efficiency and model convergence. This size allows the model to process a significant amount of data in parallel, effectively leveraging modern computational capabilities while maintaining stability in gradient updates.

1. Results Analysis and Evaluation

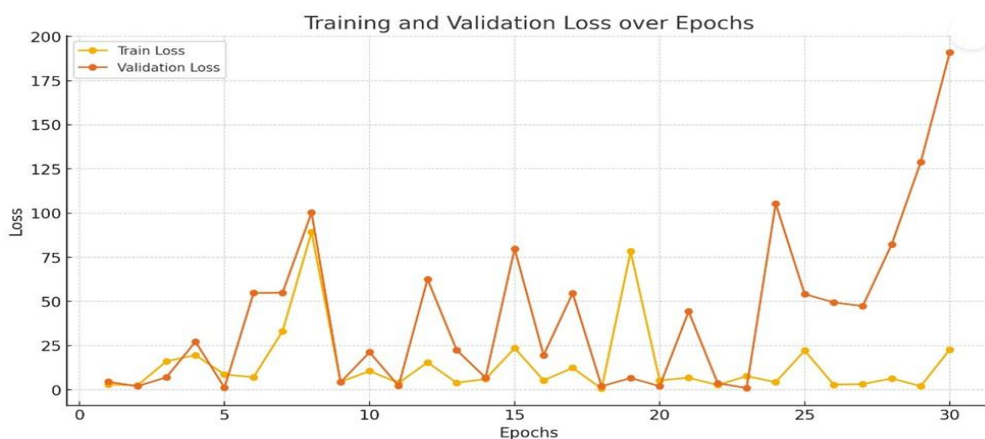
In this section, we present the results and analysis of our drug-target interaction (DTI) prediction model. To evaluate the performance of our model, we used several key metrics: training and validation loss over epochs, mean squared error (MSE), concordance index, and Pearson correlation. These metrics provide a comprehensive assessment of the model's predictive accuracy and reliability.

1. Training and Validation Loss Over Epochs:

Training and validation loss serve as pivotal indicators of a model's learning process. The training loss gauges the error on the training dataset, whereas the validation loss assesses the error on the validation dataset. Tracking these losses across epochs provides insights into the model's learning efficacy and helps in detecting signs of overfitting or underfitting. Figure 6 depicts the progression of training and validation losses throughout the epochs, offering a visual representation of the model's learning trajectory.

Figure 6 - The training and validation loss over the epochs

2. Mean Squared Error (MSE) on Validation Dataset:



The mean squared error (MSE) is a commonly employed metric for regression tasks, computing the average of the squared variances between predicted and actual values. In the realm of our DTI model, the MSE validation evaluates the alignment between predicted binding affinities and the actual values within the validation dataset.

Figure 7 displays the MSE validation outcomes, offering a glimpse into the model's precision in predicting binding affinities.

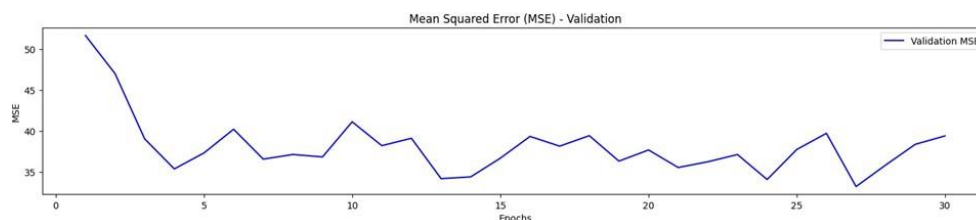


Figure 7 - The MSE validation outcomes

3. Correlation Metric:

Pearson correlation validation evaluates the linear association between predicted and actual binding affinities. Ranging from -1 to 1, a value of 1 signifies a perfect positive linear correlation, -1 represents a perfect negative linear correlation, and 0 indicates no linear correlation. A higher Pearson correlation coefficient implies that the model's predictions closely match the true binding affinities. Figure 8 showcases the Pearson correlation validation outcomes, highlighting the strength of the linear relationship between predicted and actual affinities.

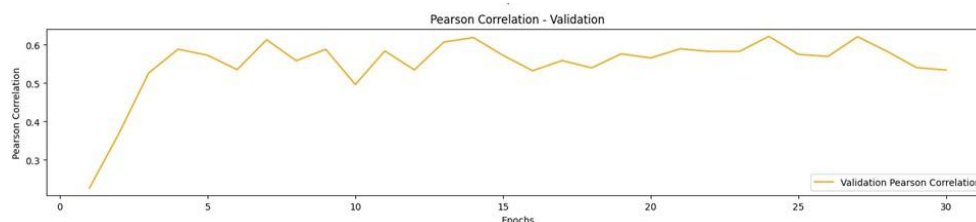


Figure 8 Pearson correlation validation outcomes

4. Concordance Index:

The concordance index (CI) is a metric utilized to assess the predictive accuracy of models, especially in survival analysis and risk prediction scenarios. In the context of our DTI model, the concordance index validation evaluates how accurately the predicted binding affinities rank the interactions in comparison to each other. A CI value nearing 1 indicates a stronger alignment between predicted and actual rankings, reflecting improved model performance. Figure 9 showcases the concordance index validation outcomes, emphasizing the model's efficacy in ranking interactions.

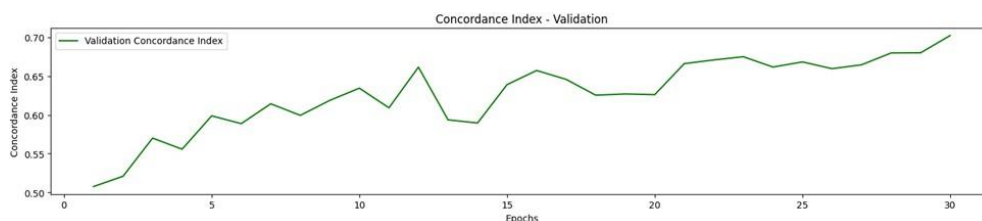


Figure 9 The concordance index validation outcomes

5. MSE on Test Dataset:

To enhance the assessment of our model's performance, we computed the mean squared error (MSE) on the test dataset at each 1000 iterations. This detailed analysis offers insights into how the model's predictive accuracy progresses throughout training. Monitoring the MSE on the test dataset is crucial for verifying that the model can generalize effectively to unseen data and does not overfit to the training dataset. Figure 10 displays the MSE on the test dataset for every 1000 iterations, shedding light on the model's performance and consistency over time.

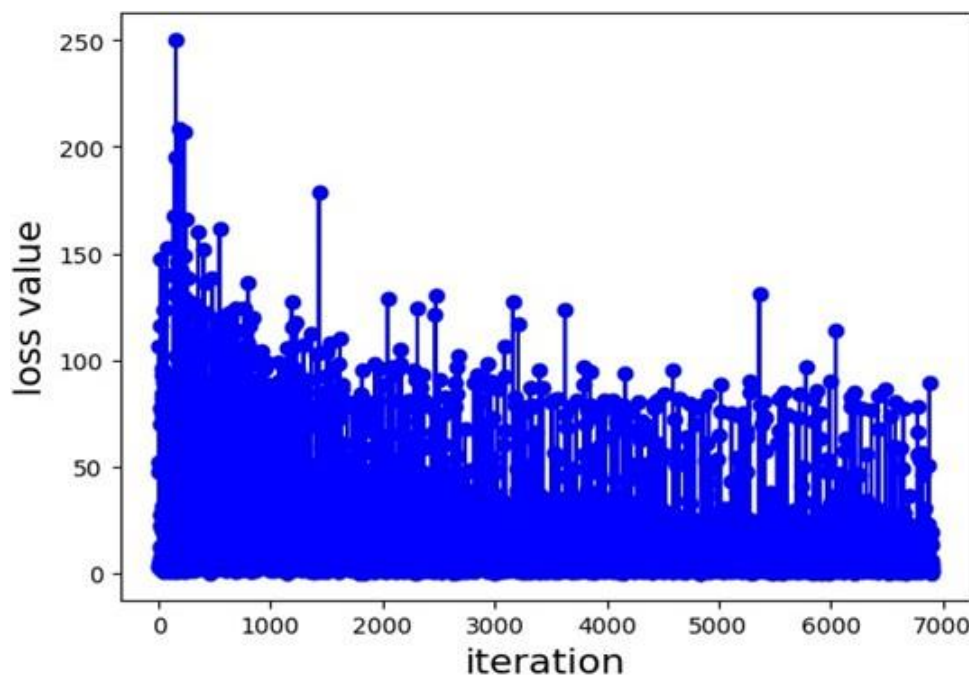


Figure 10 The MSE on the test dataset per 1000 iterations

6. Conclusion and Future Work

Our drug-target interaction prediction model excels at accurately forecasting binding affinities between various drugs and target proteins by utilizing structural data and protein sequences. Through in silico methods, this model optimizes the drug discovery process, offering insights into drug efficacy while reducing the reliance on extensive experimental data. By prioritizing candidates based on predicted affinities, researchers can efficiently allocate resources and uncover novel therapeutic options, potentially hastening drug development and improving patient outcomes.

The upcoming focus of the Drug-Target Interaction (DTI) project involves refining AI models through the integration of advanced deep learning techniques like transformers and graph neural networks. Additionally, the

inclusion of structural and omics data aims to enhance the understanding of drug-target interactions comprehensively. The goal is to broaden the model's scope to encompass new drug formats and cross-species analysis, and to ensure its practicality by incorporating clinical trial data and exploring personalized medicine applications. Future research should emphasize methodologies that amalgamate various similarities for more precise results. Efforts are directed towards identifying key molecular drivers in disease processes, highlighting the need for an improved approach to predicting DTIs due to the scarcity of knowledge regarding targeted cellular molecules by drugs. Adequate data representation is critical for effective DTI predictions, especially in Big Data scenarios characterized by sparsity and missing data, necessitating sound assumptions for imputing missing entries based on available data.

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